

# Fields and Galois Theory

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## Contents

|          |                            |          |
|----------|----------------------------|----------|
| <b>1</b> | <b>Basic definitions</b>   | <b>2</b> |
| 1.1      | Fields . . . . .           | 2        |
| 1.2      | Field extensions . . . . . | 2        |
| 1.3      | Splitting fields . . . . . | 5        |
| <b>2</b> | <b>Automorphisms</b>       | <b>7</b> |

# 1 Basic definitions

## 1.1 Fields

Recall that a *field* is a nonzero integral domain (so  $0 \neq 1$ ) in which every nonzero element is invertible; equivalently it is a commutative ring with no nontrivial ideals. The latter description of fields implies that a homomorphism of fields must be injective, since  $0 \neq 1$  implies that the kernel cannot be the entire field.

**Definition 1.1 (Characteristic of a field)** — The *characteristic* of a field  $F$  is the additive order of 1 in the field (i.e. the order of 1 in  $(F, +)$ ), or 0 if 1 does not have finite additive order.

**Exercise 1.2.** Show that the characteristic of a field is prime.

prime subfield

## 1.2 Field extensions

The reason historically for thinking about Galois theory was to talk about solutions to polynomial equations with coefficients in a particular field. The central object in this way of thinking is that of a field extension:

**Definition 1.3 (Field extension)** — If  $F$  is a field and  $K$  is a field containing  $F$ , we write that  $K/F$  (pronounced “ $K$  over  $F$ ”) is a (*field*) *extension* of  $F$ . One can also think of this as a homomorphism  $F \rightarrow K$ , since field homomorphisms are always injective.

**Example 1.4.** (1) Every field is an extension of its prime subfield.

(2) The polynomial  $x^2 + 1$  is irreducible in  $\mathbb{Q}[x]$ , so  $\mathbb{Q}[x]/(x^2 + 1)$  is a field containing  $\mathbb{Q}$ , hence is an extension of  $\mathbb{Q}$ . This is isomorphic to the subfield  $\mathbb{Q}(i) \subseteq \mathbb{C}$  generated by  $\mathbb{Q}$  and  $i$  (or a definition of the subfield, if you define  $\mathbb{C} = \mathbb{R}[x]/(x^2 + 1)$ ).

Indeed, quotients by polynomials in this way capture the idea of “adjoining a root of  $p$  to the field,” which is why field extensions contain the relevant algebraic data for trying to solve polynomials. That is, if  $F$  is a field and  $p(x) \in F[x]$  an irreducible polynomial, we can construct the extension  $F[x]/(p(x))$  of  $F$  in which  $p$  necessarily has a root  $\bar{x}$  (the image of  $x$  in the quotient).

If  $K/F$  is an extension of  $F$ , then  $K$  must be closed under addition and under multiplication by elements of  $F$ , so it is naturally an  $F$ -vector space.

**Definition 1.5 (Degree of an extension)** — The *degree* of the extension  $K/F$  is the dimension of  $K$  as an  $F$ -vector space, and it is denoted by  $[K : F]$ . If  $[K : F]$  is finite we say that the extension is *finite*.

The similarity in notation to the notation for the index of a subgroup is not an accident; we will see in fact that this notation is extremely fitting.

The other notational similarity in the definition of the degree is the choice of word: this is not a coincidence either.

**Proposition 1.6.** Let  $F$  be a field,  $p(x) \in F[x]$  an irreducible polynomial of degree  $n$ , and  $K = F[x]/(p(x))$ . Then  $[K : F] = n$  with  $F$ -basis  $1, \bar{x}, \dots, \bar{x}^{n-1}$ .

write out an example

This proposition follows from the Euclidean algorithm, where the image of a polynomial in  $F[x]$  ends up just being its remainder upon division by  $p(x)$ .

Note that the Euclidean algorithm also gives you a procedure for computing the inverse of an element in  $K$ .

**Exercise 1.7.** What is the inverse of  $\bar{x}$  in  $\mathbb{Q}[x]/(p(x))$ ? (You should be able to express your answer nicely in terms of  $p(x)$ .)

**Example 1.8.** A very, very important counterexample in Galois theory is the field extension  $\mathbb{Q}(\sqrt[3]{2}) \cong \mathbb{Q}[x]/(x^3 - 2)$ . Observe that this field does *not* contain all the roots of  $x^3 - 2$ .

Very often we want to think of an extension of  $F$  as being generated by  $F$  plus a couple of other elements from some bigger ambient field (e.g.  $\mathbb{Q}(i)$  from earlier, or  $\mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}) \subseteq \mathbb{C}$ ). Supposing that  $K/F$  is an extension of  $F$  and  $(\alpha_i)_{i \in I}$  are some elements in  $K$ , we write  $F((\alpha_i)_{i \in I}) \subseteq K$  to be the extension of  $F$  generated by the  $\alpha_i$ ; often the  $\alpha_i$  are finite in number, in which case we write  $F(\alpha_1, \dots, \alpha_n)$  and say that the extension is finitely generated. If a field  $K/F$  can be written as  $F(\alpha)$  for some  $\alpha \in K$  then we say that  $K$  is a *simple extension* of  $F$  and that  $\alpha$  is a *primitive element* of the extension.

**Proposition 1.9.** If the primitive element  $\alpha$  of a simple extension  $F(\alpha)$  is the root of a polynomial  $p(x) \in F[x]$  which is irreducible over  $F$ , then

$$F(\alpha) \cong F[x]/(p(x)).$$

The proof is a straightforward application of the first isomorphism theorem, as we have a natural homomorphism  $F[x] \rightarrow F(\alpha)$  by sending  $x$  to  $\alpha$ , whose kernel can be shown to be  $(p(x))$ . For an irreducible polynomial  $p(x) \in F[x]$ , if  $E/F$  is an extension containing a root  $\alpha$  of  $p(x)$  and  $E = F(\alpha)$ , we will call  $E$  a *stem field* of  $p(x)$ . This should be thought of as a smallest extension of  $F$  in which  $p$  has a root.

**Example 1.10.** If  $p(x) = x^3 - 2$  and  $\alpha = \sqrt[3]{2}i$ , then  $\mathbb{Q}(\alpha) \subseteq \mathbb{C}$  is isomorphic to  $\mathbb{Q}[x]/(x^3 - 2)$ , just like  $\mathbb{Q}(\sqrt[3]{2})$ , although the two extensions are obviously not equal, since one of them only has real elements. In general, for an irreducible polynomial  $p(x) \in \mathbb{Q}[x]$ , one can pick any of its roots to extend  $\mathbb{Q}$  by and still end up with an isomorphic extension, even if the various extensions are not equal. Thus, in some sense, the roots of a particular irreducible polynomial are algebraically indistinguishable, as they produce isomorphic field extensions.

We also have a converse to Proposition 1.9: if  $[F(\alpha) : F] = n$  is finite, then we can see that  $\alpha$  must satisfy a polynomial equation because the list  $(1, \alpha, \dots, \alpha^n)$  has more than  $n$  elements, hence is  $F$ -linearly dependent. By discarding factors of this polynomial we can force it to be irreducible, and then the aforementioned proposition implies that its degree is  $n$ . Thus we get the following statement:

**Proposition 1.11.** If  $F(\alpha)/F$  is a finite, simple extension, then there exists an irreducible polynomial with  $\alpha$  as a root with degree equal to  $[F(\alpha) : F]$ .

**Exercise 1.12.** If  $F(\alpha)$  is a finite simple extension of  $F$ , show that there is a unique monic irreducible polynomial  $m_{\alpha, F}(x) \in F[x]$  such that  $m_{\alpha, F}(\alpha) = 0$ . This polynomial is called the *minimal polynomial* of  $\alpha$  over  $F$ .

**Proposition 1.13.** Let  $K/F$  be an extension. If  $f(x) \in F[x]$  is a polynomial with root  $\alpha \in K$ , then  $m_{\alpha, F}(x)$  divides  $f(x)$ .

These propositions (and more to come) show that  $\alpha \in K$  being a root of a polynomial with coefficients in  $F \subseteq K$  gives us some nice information about the extension  $F(\alpha)/F$ . In this vein, we define the following notion:

**Definition 1.14** (Algebraic extension, algebraic element) — An element  $\alpha$  of the field  $K$  is *algebraic* over subfield  $F$  if  $\alpha$  is the root of some polynomial in  $F[x]$ ; equivalently (by Proposition 1.11),  $F(\alpha)/F$  is a finite simple extension. The extension  $K/F$  is an *algebraic extension* if each element of  $K$  is algebraic over  $F$ .

**Proposition 1.15.** If  $K/F$  is a finite extension, it is algebraic.

*Proof.* If  $K$  is finite-dimensional as an  $F$ -vector space, then, for any  $\alpha \in K$ , we see that  $F(\alpha) \subseteq K$ , being an  $F$ -linear subspace of  $K$ , is also finite-dimensional over  $F$ , so  $\alpha$  is algebraic over  $F$ .  $\square$

This proposition essentially just uses the inequality  $[F(\alpha) : F] \leq [K : F]$ , but can get a good amount more numerical information about degrees.

**Proposition 1.16** (Tower law). Let  $K/F$  be a subextension of the finite extension  $L/F$ , i.e.  $F \subseteq K \subseteq L$ . Then

$$[L : F] = [L : K][K : F]$$

where if one side is infinite then so is the other.

*Proof.* Let  $\alpha_1, \dots, \alpha_n$  be an  $F$ -basis for  $K$  and  $\beta_1, \dots, \beta_m$  be a  $K$ -basis for  $L$ . Then any  $x \in L$  can be written as

$$x = c_1\beta_1 + c_2\beta_2 + \dots + c_m\beta_m = \sum_{i=1}^n \sum_{j=1}^m d_{i,j} \alpha_i \beta_j$$

for  $c_i \in K$  and  $d_{i,j} \in F$ , so  $\{\alpha_i \beta_j : 1 \leq i \leq n, 1 \leq j \leq m\}$  is an  $F$ -spanning set for  $L$ . On the other hand if

$$\sum_{i=1}^n \sum_{j=1}^m d_{i,j} \alpha_i \beta_j = 0$$

then the coefficients  $\sum_{i=1}^n d_{i,j} \alpha_i$  of the  $\beta_j$  must each be zero by the linear independence of the  $\beta_j$ , and then the  $d_{i,j}$  themselves are zero by the linear independence of the  $\alpha_i$ .  $\square$

An obvious corollary of this is that for an extension  $L/F$ , any sub-extension  $K/F$  must have  $[K : F]$  dividing  $[L : F]$ . Another example of numerical information about degrees is the following:

**Proposition 1.17.** Let  $K_1/F$  and  $K_2/F$  be subextensions of the extension  $L/F$  and  $K_1 K_2$  be the smallest subfield of  $L$  containing both  $K_1$  and  $K_2$ . Then

$$[K_1 K_2 : F] \leq [K_1 : F][K_2 : F].$$

uhh just take product of bases?

*Proof.*

$\square$

**Proposition 1.18.** Let  $L/K$  and  $K/F$  be algebraic extensions. Then  $L$  is algebraic over  $F$ .

*Proof.* Let  $\alpha \in L$  and let its minimal polynomial over  $K$  be  $c_n x^n + \dots + c_1 x + c_0 \in K[x]$ . Then  $\alpha$  is an algebraic element in  $F(c_0, c_1, \dots, c_n)$ , which is a finite extension of  $F$  because the  $c_i$  are algebraic over  $F$ , so

$$[F(\alpha) : F] \leq [F(c_0, c_1, \dots, c_n, \alpha) : F] < \infty.$$

$\square$

**Remark 1.19.** A non-algebraic extension  $K/F$  is called *transcendental*. For any  $t \in K$ , the subring  $F[t]$  is isomorphic to a quotient of the polynomial ring  $F[x]$ ; if  $t$  isn't algebraic over  $F$  then we must have  $F[t] \cong F[x]$ , in which case  $t$  is *transcendental* over  $F$ .

We say that a field  $\Omega$  is *algebraically closed* if every polynomial in  $\Omega$  has a root in  $\Omega$ , or equivalently factors into a product of linear polynomials in  $\Omega$ , or equivalently that the irreducible polynomials in  $\Omega$  all have degree 1. An *algebraic closure* of field  $F$  is an algebraic extension  $\Omega/F$  for which  $\Omega$  is algebraically closed.

**Example 1.20.**  $\mathbb{C}$  is algebraically closed (fundamental theorem of algebra), and it is an algebraic closure of  $\mathbb{R}$ .

Important thing to know (not going to reproduce a proof here):

**Theorem 1.21 (Algebraic closures exist).** Every field  $F$  has an algebraic closure, and any two algebraic closures of  $F$  are isomorphic (though not at all uniquely).

### 1.3 Splitting fields

The key properties we will establish about field extensions are all combinatorial in flavor, since finite-degree extensions are fairly rigid objects.

**Proposition 1.22.** Let  $F(\alpha)/F$  be a simple extension, and  $\varphi_0: F \rightarrow \Omega$  a homomorphism to some field  $\Omega$ .

- (a) If  $\alpha$  is transcendental over  $F$ , there is a one-to-one correspondence between homomorphisms  $F(\alpha) \rightarrow \Omega$  extending  $\varphi_0$  and transcendental elements of  $\Omega$ .
- (b) If  $\alpha$  is algebraic over  $F$  with minimal polynomial  $m(x) \in F[x]$ , there is a one-to-one correspondence between homomorphisms  $F(\alpha) \rightarrow \Omega$  extending  $\varphi_0$  and roots of  $\varphi_0 m$  in  $\Omega$ , where  $\varphi_0 m(x) \in F'[x]$  is the polynomial obtained by applying  $\varphi_0$  to the coefficients of  $m$ .

*Proof.*

□

**Definition 1.23 (Splitting field)** — Let  $E/F$  be a field extension and  $f(x) \in F[x]$ . We say that  $f$  *splits in  $E$*  or  $E$  *splits  $f$*  if  $f$  factors into a product of linear polynomials in  $E[x]$ , i.e.

$$f(x) = c(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n), \quad c \in F, \alpha_i \in E.$$

A field  $E$  in which  $f$  splits is a *splitting field* for  $f$  if  $E$  is also generated by the roots of  $f$  in  $E$ .

**Example 1.24.** (1) If  $f(x) = ax^2 + bx + c \in \mathbb{Q}[x]$  does not have rational roots, then  $\mathbb{Q}(\sqrt{b^2 - 4ac})$  is a splitting field for  $f$ .

(2) Let  $f(x) \in \mathbb{Q}[x]$  be an irreducible cubic polynomial with roots  $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{C}$ . Then  $\mathbb{Q}(\alpha_1, \alpha_2) \subseteq \mathbb{C}$  certainly is a splitting field for  $f$  (because we can recover  $\alpha_3$  by Vieta's formulas). In  $\mathbb{Q}(\alpha_1)$  we can factor  $f(x) = (x - \alpha_1)g(x)$  for some degree 2 polynomial  $g(x)$ , so, depending on the irreducibility of  $g$  in  $\mathbb{Q}(\alpha_1)$ , we get that the splitting field  $\mathbb{Q}(\alpha_1, \alpha_2)$  either has degree 3 or 6 over  $\mathbb{Q}$  by the tower law.

(3) Let  $f(x) = 1 + x + \cdots + x^{p-1} = \frac{x^p - 1}{x - 1} \in \mathbb{Q}[x]$ . If complex number  $\zeta \neq 1$  satisfies  $\zeta^p = 1$ , then the roots of  $f$  are  $\zeta, \zeta^2, \dots, \zeta^{p-1}$ , so  $\mathbb{Q}[\zeta]$  is a splitting field for  $f$  with degree  $p - 1$ .

(4) Let  $f(x) = x^p - x - a \in F[x]$ , where  $F$  has characteristic  $p \neq 0$ . Since  $r^p = r$  for all  $r$  in the prime subfield  $\mathbb{F}_p$ , if  $\alpha$  is a root of  $f(x)$  in some extension  $E$ , then so are  $\alpha + 1, \dots, \alpha + p - 1$ , meaning  $F(\alpha)$  is a splitting field for  $f$ .

In (2) of Example 1.24, we used the fact that  $\mathbb{C}$  is algebraically closed to obtain a splitting field for  $f$  (and we can do this in general for any polynomial in  $\mathbb{Q}[x]$ ). In general, splitting fields always exist:

**Proposition 1.25.** Let  $F$  be a field and  $f(x) \in F[x]$  a polynomial. A splitting field  $E/F$  of  $f$

exists, and  $[E : F] \leq (\deg f)!$ .

*Proof.* Let  $f_0 = f$  and  $F_0 = F$ , and generate a sequence of polynomials and extensions of  $F$  as follows: let  $F_{n+1} = F_n(\alpha_n)$  be a stem field for an irreducible factor of the polynomial  $f_n$  (i.e.  $F_{n+1}$  contains a root  $\alpha_n$  of this factor and is generated by this root over  $F_n$ ), and let  $f_{n+1}(x) = f_n(x)/(x - \alpha_n)$ . Clearly  $E = F_{\deg f}$  is a splitting field for  $f$ . Since  $\deg f_{n+1} = \deg f_n - 1$ ,  $\deg f_n = \deg f - n$ , so  $[F_{n+1} : F_n] \leq \deg f - n$ , and the tower law says that

$$[E : F] = \prod_{n=1}^{\deg f} [F_n : F_{n-1}] \leq \prod_{n=1}^{\deg f} (\deg f - n) = (\deg f)!. \quad \square$$

The main question now about splitting fields is how much leeway we have in their construction; for example, to what extent do the particular sequences  $F_n$  and  $f_n$  used to generate a splitting field in Proposition 1.25 change the field?

**Lemma 1.26.** Let  $F$  be a field and  $f(x) \in F[x]$ . Let  $E/F$  and  $\Omega/F$  be extensions with  $E$  generated by the roots of  $f$  in  $E$  and  $\Omega$  splitting  $f$ . There exists an  $F$ -homomorphism  $\varphi: E \rightarrow \Omega$ , and the number of such  $F$ -homomorphisms is bounded above by  $[E : F]$ , with equality if  $f$  has distinct roots in  $\Omega$ .

*Proof.* Let  $\alpha_1, \dots, \alpha_m$  be the roots of  $f$  in  $E$  and define  $F_k = F(\alpha_1, \dots, \alpha_k)$  for  $1 \leq k \leq m$  (so  $F_m = E$ ), and  $F_0 = F$ . Furthermore, let  $f_k$  be the minimal polynomial of  $\alpha_{k+1}$  in  $F_k$  for  $0 \leq k < m$ , so  $f_0 = f$ . □

**Corollary 1.27.** Let  $F$  be a field and  $f(x) \in F[x]$ . Any two splitting fields  $E/F$  and  $E'/F$  are  $F$ -isomorphic.

*Proof.* There exist  $F$ -homomorphisms  $E \rightarrow E'$  and  $E' \rightarrow E$ , which respectively imply that  $[E : F] \leq [E' : F]$  and  $[E' : F] \leq [E : F]$ . Thus the two extensions have the same finite degree over  $F$ , so dimension counting implies that the  $F$ -homomorphisms are linear isomorphisms, which implies that they are bijective, hence also  $F$ -isomorphisms. □

Note that there is no canonical choice of isomorphism between two splitting fields, so one still has to be careful when talking about “the splitting field” of a polynomial. One thing we can say about polynomials now is that their factorizations in their splitting fields are essentially unique. Suppose that  $F$  is a field,  $f(x) \in F[x]$ , and  $E/F$  and  $E'/F$  are splitting fields for  $f$  in which

$$f(x) = c(x - \alpha_1) \cdots (x - \alpha_n) = c(x - \beta_1) \cdots (x - \beta_n), \quad c \in F, \alpha_i \in E, \beta_i \in E'$$

(note of course that the number of  $\alpha_i$ s and  $\beta_i$ s is the same, namely equal to the degree of  $f$ ). Then, an  $F$ -isomorphism  $\varphi: E \rightarrow E'$  induces an isomorphism of the respective polynomial rings  $\varphi: E[x] \rightarrow E'[x]$  by just sending  $x$  to  $x$ , in which

$$\varphi f(x) = c \prod_{i=1}^n (x - \varphi(\alpha_i)),$$

so unique factorization implies that the  $\varphi(\alpha_i)$  are a permutation of the  $\beta_i$  (as multisets). In particular, the multiplicities of the roots of a polynomial in a splitting field are intrinsic to the polynomial and independent of the choice of extension (as we should hope).

**Definition 1.28 (Separable polynomial)** — Let  $F$  be a field. A polynomial  $f(x) \in F[x]$  is *separable* if it has no multiple roots in any splitting field, and inseparable otherwise.

There is an easy way to check whether a polynomial is separable. Define the *formal derivative* of  $f(x) = a_0 + a_1x + \cdots + a_nx^n$  to be the expected expression:

$$f'(x) = a_1 + 2a_2x + \cdots + na_nx^{n-1}.$$

This is a derivation on  $F[x]$  (i.e. is an  $F$ -linear map that satisfies the product rule). The product rule implies that a root of  $f(x)$  with multiplicity  $m$  not divisible by the characteristic  $p$  of  $F$  (of course, this requires  $p \neq 0$ ) is also a root of  $f'(x)$  with multiplicity  $m - 1$ , since if  $f(x) = (x - \alpha)^m g(x)$  for some polynomial  $g(x)$  with  $g(\alpha) \neq 0$ , then

$$f'(x) = (x - \alpha)^m g'(x) + m(x - \alpha)^{m-1} g(x) = (x - \alpha)^{m-1} ((x - \alpha)g'(x) + mg(x))$$

and  $(x - \alpha)g'(\alpha) + mg(\alpha)$  is clearly nonzero at 0 unless  $p$  divides  $m$ . If  $m$  is divisible by  $p$  then  $f'(x) = (x - \alpha)^m g'(x)$ , so  $\alpha$  is a root of  $f'$  with multiplicity at least  $m$ . Either way, if  $f$  has a multiple root, then its derivative also has that root (and you can keep track of its multiplicity, which is especially nice in characteristic zero). Conversely if  $f$  does not have any multiple roots then  $m = 1$  above, which is not divisible by any possible value of the characteristic.

**Proposition 1.29.** A polynomial  $f(x) \in F[x]$  is separable if and only if  $f$  and  $f'$  are relatively prime.

This is fantastic, since derivatives are easy to compute by just looking at the coefficients and the greatest common divisor is easy to compute via the Euclidean algorithm.

**Exercise 1.30.** Let  $f(x), g(x) \in F[x]$  and let  $E/F$  be an extension. Show that the greatest common divisor of  $f$  and  $g$  in  $F[x]$  is the same as the greatest common divisor in  $E[x]$ . Thus, the greatest common divisor of  $f$  and  $f'$  is independent of the ambient field.

**Example 1.31.** (1) Any irreducible polynomial over a characteristic zero field is separable, as are products of distinct irreducible polynomials.

(2) Let  $a$  be an element of the field  $F$  with characteristic  $p$ . The polynomial  $x^p - x - a$  is separable, and it is irreducible if for example  $a = 1$  (see Example 1.24).

(3) A field  $F$  with characteristic  $p \neq 0$  is *perfect* if  $F^p := \{x^p : x \in F\}$  is equal to  $F$ . Then any irreducible polynomial in  $F[x]$  is separable. If  $f$  were inseparable, then  $\gcd(f, f') \neq 1$  and the irreducibility of  $f$  implies that  $f$  divides  $f'$ , which forces  $f' = 0$  since  $\deg f' < \deg f$ . Thus  $f(x) = g(x^p) = g(x)^p$  for some polynomial  $g(x) \in F[x]$  because the coefficients of  $g$  are  $p$ th powers, contradicting irreducibility.

(4) If  $F$  isn't a perfect field and  $a \in F$  is not a  $p$ th power then  $x^p - a$  is irreducible and inseparable, since if  $\alpha^p = a$  then  $x^p - a = (x - \alpha)^p$  so  $\alpha$  is the only root of the polynomial and has multiplicity  $p$ . In fact, this scenario, where the field is imperfect and the derivative is zero, is the only obstruction to an irreducible polynomial being inseparable.

## 2 Automorphisms

**Definition 2.1 (Automorphism)** — The group  $\text{Aut}(E/F)$  is the group of  $F$ -automorphisms, which are  $F$ -isomorphisms  $E \rightarrow E$ .

**Example 2.2.** The conjugation map  $\bar{\cdot}: \mathbb{C} \rightarrow \mathbb{C}$  is an  $\mathbb{R}$ -automorphism of  $\mathbb{C}$ .